6.8 Quiz: Polynomials — Markscheme (b)

1. [Maximum mark: 8]

   Let $f(x) = x^2 + 6x - 16$.

   (a) Write $f(x)$ in factored form $(x - a)(x - b)$. [2]

   $f(x) = (x - 2)(x - (-8)) = (x - 2)(x + 8)$ **A1A1**

   Accept $(x + 8)(x - 2)$.

   (b) Write down the x-intercepts. [1]

   $x = 2$ and $x = -8$ **A1**

   Accept $(2, 0)$ *and* $(-8, 0)$.

   FT: award A1 for x-intercepts read from the candidate's factors in (a).

   (c) Write down the y-intercept. [1]

   $f(0) = -16$, so the y-intercept is $(0, -16)$. **A1**

   Accept -16.

   FT: award A1 for y-intercept correctly computed from the candidate's factored form in (a).

   (d) Write $f(x)$ in vertex form $(x - h)^2 + k$. [3]

   Axis of symmetry: $x = \dfrac{-8 + 2}{2} = -3$ **M1**

   $f(x) = (x + 3)^2 - 25$ **A1A1**

   Accept $h = -3,\ k = -25$.

   M1 may also be awarded for any attempt at completing the square.

   (e) Write down the coordinates of the vertex. [1]

   Vertex: $(-3, -25)$ **A1**

   FT: award A1 for vertex coordinates read from the candidate's vertex form in (d).

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2. [Maximum mark: 7]

   Consider the function $g(x) = -(x - 5)^2 + 4$.

   (a) Write down the coordinates of the vertex. [2]

   Vertex: $(5, 4)$ **A1A1**

(b) Write down the equation of the axis of symmetry. [1]

$x = 5$ **A1**

(c) Show that $g(x) = -x^2 + 10x - 21$. [2]

$g(x) = -(x-5)^2 + 4 = -(x^2 - 10x + 25) + 4$ **M1**

$= -x^2 + 10x - 25 + 4 = -x^2 + 10x - 21$ **A1 AG**

(d) Solve $g(x) = 0$ to find the x-intercepts. [2]

$-x^2 + 10x - 21 = 0$

$x^2 - 10x + 21 = 0$

$(x-3)(x-7) = 0$

$x = 3$ or $x = 7$ **A1A1**

3. [Maximum mark: 6]

The line $y = 4x$ and the parabola $y = x^2 - 5$.

(a) Show that the x-coordinates of intersection satisfy $x^2 - 4x - 5 = 0$. [1]

At intersection: $x^2 - 5 = 4x$ **M1**

$x^2 - 4x - 5 = 0$ **AG**

(b) Show that $x^2 - 4x - 5 = 0$ has two distinct real roots. [1]

$\Delta = (-4)^2 - 4(1)(-5) = 16 + 20 = 36 > 0$ **M1**

Since $\Delta > 0$, there are two distinct real roots.

(c) Factorise $x^2 - 4x - 5$. [1]

$(x - 5)(x + 1)$ **A1**

(d) Find the x-coordinates of the points of intersection. [1]

$x = 5$ and $x = -1$ **A1**

FT: award A1 for x-coordinates read from the candidate's factors in (c).

(e) Find the coordinates of both points of intersection. [2]

$x = 5$: $y = 4(5) = 20$, so $(5, 20)$ **A1**

$x = -1$: $y = 4(-1) = -4$, so $(-1, -4)$ **A1**

FT: award A1 for each y-value correctly computed from the candidate's x-value.

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4. [Maximum mark: 7]

The equation $2x^2 - kx + 32 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k. [1]

$\Delta = (-k)^2 - 4(2)(32) = k^2 - 256$ **M1**

(b) Find the values of k. [2]

Equal roots: $\Delta = 0$ **M1**

$k^2 - 256 = 0$

$k^2 = 256$

$k = \pm 16$ **A1**

(c) For $k = 16$, find the repeated root. [2]

$2x^2 - 16x + 32 = 0$

$x^2 - 8x + 16 = 0$ **M1**

$(x - 4)^2 = 0$

$x = 4$ **A1**

(d) State the values of k for which the equation has no real roots. [2]

$\Delta < 0$ **M1**

$k^2 - 256 < 0$

$k^2 < 256$

$-16 < k < 16$ **A1**

Accept $|k| < 16$.